LOCAL EXISTENCE, UNIQUENESS, OSGOOD'S THEOREM, AND MAXIMAL INTERVALS FOR FIRST-ORDER ODE SYSTEMS

Contents

1. Setup and notation

We study the initial value problem (IVP)

$$x'(t) = f(t, x(t)), \qquad x(t_0) = x_0, \tag{1}$$

where $x(\cdot) : I \to \mathbb{R}^n$, $I \subset \mathbb{R}$ an interval with $t_0 \in I$, and $f : D \to \mathbb{R}^n$ is a given function on an open set $D \subset \mathbb{R} \times \mathbb{R}^n$ with $(t_0, x_0) \in D$. A (classical) *solution on I* is a C^1 map $x : I \to \mathbb{R}^n$ such that $(t, x(t)) \in D$ for $t \in I$ and (1) holds, including the initial condition.

The integral form equivalent to (1) is

$$x(t) = x_0 + \int_{t_0}^{t} f(s, x(s))\, ds. \tag{2}$$

We write $\|\cdot\|$ for any norm on $\mathbb{R}^n$ (all norms are equivalent in finite dimensions). For $X = C([a, b]; \mathbb{R}^n)$ we use the sup norm $\|x\|_\infty = \sup_{t\in[a,b]} \|x(t)\|$.

Lemma 1.1 (Completeness of the sup-norm space). *$C([a, b]; \mathbb{R}^n)$ with $\|\cdot\|_\infty$ is a Banach space: every Cauchy sequence converges uniformly to a continuous limit.*

Proof. A uniform limit of continuous functions is continuous. The completeness of $(\mathbb{R}^n, \|\cdot\|)$ and standard ε–N arguments yield completeness of $C([a, b]; \mathbb{R}^n)$. $\square$

2. Banach's contraction principle

Definition 2.1 (Contraction). A map $T : (X, d) \to (X, d)$ on a metric space is a *contraction* if there exists $q \in [0, 1)$ such that $d(Tu, Tv) \le q\, d(u, v)$ for all $u, v \in X$.

Theorem 2.2 (Banach fixed-point). *Let (X, d) be complete and $T : X \to X$ a contraction with constant $q \in [0, 1)$. Then:*

(i) There is a unique fixed point x^* with $Tx^* = x^*$.

(ii) *For any* $x_0 \in X$, *the iterates* $x_{k+1} = Tx_k$ *converge to* x^*.

(iii) *Error estimate:* $d(x_k, x^*) \le \dfrac{q^k}{1-q}\, d(x_1, x_0)$.

Proof. Define $x_{k+1} = Tx_k$. Then

$$d(x_{k+1}, x_k) \le q\, d(x_k, x_{k-1}) \le \cdots \le q^k d(x_1, x_0).$$

Hence for $m > k$,

$$d(x_m, x_k) \le \sum_{j=k}^{m-1} d(x_{j+1}, x_j) \le d(x_1, x_0) \sum_{j=k}^{m-1} q^j = d(x_1, x_0)\, \frac{q^k(1-q^{m-k})}{1-q}.$$

Thus (x_k) is Cauchy; completeness yields $x_k \to x^* \in X$. Continuity of T (automatic for contractions) gives $Tx^* = \lim Tx_k = \lim x_{k+1} = x^*$. If $Ty = y$, then $d(y, x^*) = d(Ty, Tx^*) \le q\, d(y, x^*)$, so $d(y, x^*) = 0$ and $y = x^*$. The error bound follows from the telescoping estimate above and the triangle inequality:

$$d(x_k, x^*) \le \sum_{j=k}^{\infty} d(x_{j+1}, x_j) \le \frac{q^k}{1-q}\, d(x_1, x_0).$$

□

3. Grönwall-type inequalities

3.1. Integral Grönwall.

Lemma 3.1 (Integral Grönwall). *Let* $u : [t_0, T] \to [0, \infty)$ *be continuous,* $A \ge 0$, *and* $B : [t_0, T] \to [0, \infty)$ *continuous. If*

$$u(t) \le A + \int_{t_0}^{t} B(s)\, u(s)\, ds \qquad (t \in [t_0, T]),$$

then

$$u(t) \le A \exp\Big(\int_{t_0}^{t} B(s)\, ds\Big) \qquad (t \in [t_0, T]).$$

Proof. Define $v(t) := A + \int_{t_0}^{t} B(s)\, u(s)\, ds$. Then $u \le v$ and v is absolutely continuous with $v'(t) = B(t)\, u(t) \le B(t)\, v(t)$ almost everywhere. Consider $w(t) := v(t) \exp(-\int_{t_0}^{t} B)$. Then $w'(t) = e^{-\int B}(v'(t) - B(t)v(t)) \le 0$ a.e., so $w(t) \le w(t_0) = A$. Therefore

$$u(t) \le v(t) \le A\, \exp\Big(\int_{t_0}^{t} B(s)\, ds\Big).$$

□

3.2. Differential Grönwall with forcing.

Lemma 3.2 (Differential Grönwall). *Let* $z : [t_0, T] \to [0, \infty)$ *be absolutely continuous,* $b, f : [t_0, T] \to [0, \infty)$ *integrable, and assume*

$$z'(t) \le b(t)\, z(t) + f(t) \quad \text{for a.e. } t \in [t_0, T].$$

Then

$$z(t) \le z(t_0)\, e^{\int_{t_0}^{t} b} + \int_{t_0}^{t} e^{\int_{s}^{t} b(\tau)\, d\tau}\, f(s)\, ds.$$

In particular, if $f \equiv 0$ *then* $z(t) \le z(t_0)\, e^{\int_{t_0}^{t} b}$.

Proof. Set $\beta(t) = \int_{t_0}^{t} b$. The integrating factor gives

$$\frac{d}{dt}(e^{-\beta(t)} z(t)) = e^{-\beta(t)}(z'(t) - b(t)z(t)) \le e^{-\beta(t)} f(t).$$

Integrate from t_0 to t to obtain the stated bound. □

4. Picard–Lindelöf/Cauchy–Lipschitz

We prove existence, uniqueness, an explicit local existence time, and continuous dependence.

Theorem 4.1 (Picard–Lindelöf)**.** *Suppose* $(t_0, x_0) \in D$*, and there is a* rectangle

$$\mathcal{R} = [t_0 - a,\, t_0 + a] \times \overline{B_R(x_0)} \subset D$$

such that

(a) f *is continuous on* $\mathcal{R}$ *and bounded by* M*, i.e.* $\|f(t,x)\| \le M$ *on* $\mathcal{R}$*;*
(b) f *is Lipschitz in* x *on* $\mathcal{R}$ *with constant* $L \ge 0$*:* $\|f(t,x) - f(t,y)\| \le L\,\|x-y\|$ *for all* $(t,x),(t,y) \in \mathcal{R}$*.*

Choose

$$T \;:=\; \min\Big\{\, a,\ \frac{R}{M},\ \frac{1}{2L} \Big\} \quad \text{with the convention } \tfrac{1}{2L} = +\infty \text{ if } L = 0.$$

Then there is a unique solution $x : [t_0 - T, t_0 + T] \to \mathbb{R}^n$ *to* (1)*. Moreover, the Picard iterates*

$$x^{(0)}(t) \equiv x_0, \qquad x^{(k+1)}(t) = x_0 + \int_{t_0}^{t} f(s, x^{(k)}(s))\, ds$$

converge uniformly to x *with the error estimate*

$$\left\| x^{(k)} - x \right\|_\infty \;\le\; \frac{q^k}{1-q}\, \left\| x^{(1)} - x^{(0)} \right\|_\infty, \quad q := LT \le \tfrac{1}{2}.$$

Proof. Work on $I = [t_0 - T, t_0 + T]$ and the complete metric space $C(I;\mathbb{R}^n)$ with $\|\cdot\|_\infty$. Define the closed, hence complete, subset

$$X := \Big\{ x \in C(I;\mathbb{R}^n) : \sup_{t \in I} \|x(t) - x_0\| \le R \Big\}.$$

Define the Picard operator $\Gamma : X \to C(I;\mathbb{R}^n)$ by

$$(\Gamma x)(t) = x_0 + \int_{t_0}^{t} f(s, x(s))\, ds.$$

Step 1 (self-map). For $x \in X$ and $t \in I$,

$$\|(\Gamma x)(t) - x_0\| \le \int_{t_0}^{t} \|f(s,x(s))\|\, ds \le M\,|t - t_0| \le MT \le R.$$

Thus $\Gamma(X) \subset X$.

Step 2 (contraction). For $x, y \in X$ and $t \in I$,

$$\|(\Gamma x)(t) - (\Gamma y)(t)\| \le \int_{t_0}^{t} \|f(s,x(s)) - f(s,y(s))\|\, ds \le L \int_{t_0}^{t} \|x(s) - y(s)\|\, ds \le LT\, \|x - y\|_\infty .$$

Hence $\|\Gamma x - \Gamma y\|_\infty \le q\,\|x - y\|_\infty$ with $q = LT \le \frac{1}{2}$.

Step 3 (fixed point). By Banach (Theorem 2.2), Γ has a unique fixed point $x \in X$, which satisfies (2) and hence (1). The error estimate follows from Theorem 2.2(iii). □

Corollary 4.2 (Uniqueness via Grönwall)**.** *Under the hypotheses of Theorem 4.1, the solution on* I *is unique: if* x, y *solve* (1) *on* I*, then* $x \equiv y$*.*

Proof. Subtract the integral equations:

$$\|x(t) - y(t)\| \le \int_{t_0}^{t} \|f(s,x(s)) - f(s,y(s))\|\, ds \le L \int_{t_0}^{t} \|x(s) - y(s)\|\, ds.$$

Apply Lemma 3.1 with $A = 0$ and $B \equiv L$ to conclude $\|x(t) - y(t)\| \equiv 0$. □

Proposition 4.3 (Continuous dependence on initial data)**.** *Under the hypotheses of Theorem 4.1, let* x *and* y *solve* $x' = f(t,x)$ *on* I *with initial data* $x(t_0) = x_0$ *and* $y(t_0) = y_0$*. Then for* $t \in I$*,*

$$\|x(t) - y(t)\| \le e^{L|t - t_0|}\, \|x_0 - y_0\| .$$

Proof. From the integral equations,

$$\|x(t) - y(t)\| \le \|x_0 - y_0\| + L \int_{t_0}^{t} \|x(s) - y(s)\| \, ds.$$

Apply Lemma 3.1 with $A = \|x_0 - y_0\|$ and $B \equiv L$. □

5. One-sided Lipschitz (OSL) uniqueness

Definition 5.1 (One-sided Lipschitz)**.** We say f is *one-sided Lipschitz* with constant $L \in \mathbb{R}$ on a region $\Omega \subset D$ if

$$\langle f(t,x) - f(t,y),\, x - y \rangle \le L\, \|x - y\|^2 \quad \text{for all } (t,x), (t,y) \in \Omega.$$

Theorem 5.2 (Uniqueness under OSL)**.** *Assume f is continuous and locally Lipschitz*[1] *in x (for existence), and satisfies the OSL condition (5.1) on a region containing two solutions x, y with the same initial data. Then $x \equiv y$ on their common interval of existence.*

Proof. Let $w(t) = \|x(t) - y(t)\|^2$. Then

$$\frac{1}{2} w'(t) = \langle x'(t) - y'(t),\, x(t) - y(t) \rangle = \langle f(t, x(t)) - f(t, y(t)),\, x(t) - y(t) \rangle \le L\, w(t).$$

By Lemma 3.2 with $f \equiv 0$, we get $w(t) \le w(t_0) e^{2L(t - t_0)}$. Since $w(t_0) = 0$, $w \equiv 0$. □

6. Bihari–LaSalle inequality (full proof) and Osgood uniqueness

6.1. A tool: upper right Dini derivative and a comparison lemma.

For a continuous function $h : [t_0, T] \to \mathbb{R}$, the *upper right Dini derivative* at t is

$$D^+ h(t) := \limsup_{h \downarrow 0} \frac{h(t+h) - h(t)}{h}.$$

We use the following standard comparison lemma.

Lemma 6.1 (Dini derivative comparison)**.** *Let $w : [t_0, T] \to \mathbb{R}$ be continuous and $g : [t_0, T] \to \mathbb{R}$ integrable. If*

$$D^+ w(t) \le g(t) \quad \textit{for all } t \in [t_0, T),$$

then $w(t) \le w(t_0) + \int_{t_0}^{t} g(s)\, ds$ for all $t \in [t_0, T]$.

Proof. Suppose not. Then there exists t_1 with

$$\eta := \sup_{t \in [t_0, T]} \left(w(t) - w(t_0) - \int_{t_0}^{t} g \right) > 0.$$

Let τ be the first time such that $w(\tau) - w(t_0) - \int_{t_0}^{\tau} g = \eta$. By minimality of τ and continuity, $D^+ w(\tau) \ge g(\tau)$. This contradicts $D^+ w(\tau) \le g(\tau)$. □

6.2. Bihari–LaSalle inequality (continuous g case).

Lemma 6.2 (Bihari–LaSalle)**.** *Let $u : [t_0, T] \to [0, \infty)$ be continuous and $g : [t_0, T] \to [0, \infty)$ continuous. Let $\phi : [0, \rho) \to (0, \infty)$ be continuous and nondecreasing. Suppose*

$$u(t) \le a + \int_{t_0}^{t} g(s)\, \phi(u(s))\, ds \qquad (t \in [t_0, T]), \tag{3}$$

with $a \in [0, \rho)$. Define

$$\Phi(r) := \int_{r_*}^{r} \frac{ds}{\phi(s)} \quad \textit{for any } r_* \in (0, \rho),$$

[1] If every point has a neighborhood U so that $f|_U$ is Lipschitz with a Lipschitz constant that is (a priori) dependent on U.

so that Φ is C^1 and strictly increasing on $(0,\rho)$. Then

$$\Phi(u(t)) \le \Phi(a) + \int_{t_0}^{t} g(s)\,ds \qquad (t \in [t_0,T]).$$

In particular, if $a=0$ and $\displaystyle\int_{0+} \frac{ds}{\phi(s)} = \infty$, then $u(t)\equiv 0$.

Proof. Fix $\varepsilon \in (0,\rho-a)$. Set $u_\varepsilon(t) := u(t)+\varepsilon$. Since ϕ is nondecreasing and $u \le u_\varepsilon$,

$$u_\varepsilon(t) \le a+\varepsilon+\int_{t_0}^{t} g(s)\,\phi(u_\varepsilon(s))\,ds.$$

For $h>0$, subtract the same inequality at time t from that at $t+h$ to get

$$u_\varepsilon(t+h)-u_\varepsilon(t) \le \int_{t}^{t+h} g(s)\,\phi(u_\varepsilon(s))\,ds.$$

Divide by h and let $h \downarrow 0$. By continuity of g and $\phi \circ u_\varepsilon$,

$$D^+u_\varepsilon(t) \le g(t)\,\phi(u_\varepsilon(t)).$$

Let Φ be as in the statement. For $h>0$,

$$\frac{\Phi(u_\varepsilon(t+h))-\Phi(u_\varepsilon(t))}{h} = \frac{1}{h}\int_{u_\varepsilon(t)}^{u_\varepsilon(t+h)} \frac{ds}{\phi(s)} \le \frac{u_\varepsilon(t+h)-u_\varepsilon(t)}{h}\cdot\frac{1}{\phi(u_\varepsilon(t))},$$

since $s \mapsto 1/\phi(s)$ is nonincreasing. Taking $\limsup$ gives

$$D^+(\Phi\circ u_\varepsilon)(t) \le \frac{1}{\phi(u_\varepsilon(t))}\,D^+u_\varepsilon(t) \le g(t).$$

By Lemma 6.1,

$$\Phi(u_\varepsilon(t)) \le \Phi(u_\varepsilon(t_0)) + \int_{t_0}^{t} g(s)\,ds = \Phi(a+\varepsilon)+\int_{t_0}^{t} g(s)\,ds.$$

Let $\varepsilon \downarrow 0$ and use continuity of Φ on $(0,\rho)$ to obtain the claim. If $a=0$ and $\int_{0+} ds/\phi(s)=\infty$, then $\Phi(0^+)=-\infty$; the inequality forces $u(t)=0$. ☐

Theorem 6.3 (Osgood uniqueness)**.** *Let f be continuous on a region containing the trajectories, and suppose there exists a* modulus of continuity *$\omega : [0,\delta) \to [0,\infty)$ (continuous, increasing, $\omega(0)=0$) such that*

$$\|f(t,x)-f(t,y)\| \le \omega(\,\|x-y\|\,) \qquad \textit{whenever defined.}$$

If $\displaystyle\int_{0+}\frac{dr}{\omega(r)} = \infty$ (Osgood condition), then solutions to (1) are unique.

Proof. Let x,y be two solutions with $x(t_0)=y(t_0)=x_0$ on a common interval $[t_0,T]$. Set $u(t)=\|x(t)-y(t)\|$. From the integral equation,

$$u(t) \le \int_{t_0}^{t} \omega(u(s))\,ds.$$

Apply Lemma 6.2 with $a=0$, $g\equiv 1$, and $\phi=\omega$. The divergence $\int_0 dr/\omega(r)=\infty$ forces $u(t)\equiv 0$. ☐

Remark 6.4. Lipschitz continuity corresponds to $\omega(r)=Lr$ and satisfies Osgood; Hölder moduli $\omega(r)=Cr^\alpha$ with $\alpha\in(0,1)$ fail Osgood and uniqueness may fail.

7. Continuation, maximal intervals, and blow-up

Lemma 7.1 (Continuation/extension)**.** *Let f be continuous and locally Lipschitz in x on D. Let $x : (\alpha, \beta) \to \mathbb{R}^n$ be a solution to (1) with $(t, x(t)) \in D$ for all $t \in (\alpha, \beta)$ and $t_0 \in (\alpha, \beta)$. If $\beta < \infty$ and there exists a compact $K \subset \mathbb{R}^n$ such that*

$$\{(t, x(t)) : t \in [t_0, \beta)\} \subset [t_0, \beta] \times K \subset D,$$

then $x(t)$ extends continuously to $t = \beta$ and the solution extends past β.

Proof. By continuity of f on $[t_0, \beta] \times K$, there is M with $\|f\| \le M$. For $t, s \in [t_0, \beta)$,

$$\|x(t) - x(s)\| = \left\| \int_s^t f(\tau, x(\tau))\, d\tau \right\| \le M|t - s|.$$

Thus x is Cauchy as $t \uparrow \beta$ and has a limit $x_\beta \in K$; define $x(\beta) = x_\beta$. Local existence (Theorem 4.1) at (β, x_β) yields an extension to $t > \beta$. □

Lemma 7.2 (Vanishing-interval average of a continuous map)**.** *Let $g : [a, b] \to \mathbb{R}^m$ be continuous. Then*

$$\lim_{t \uparrow b} \frac{1}{b - t} \int_t^b g(s)\, ds \;=\; g(b).$$

Proof. By continuity, g is uniformly continuous on $[a, b]$. Given $\varepsilon > 0$, pick $\delta > 0$ such that $\|g(s) - g(b)\| < \varepsilon$ whenever $s \in [b - \delta, b]$. For $t \in (b - \delta, b)$,

$$\left\| \frac{1}{b - t} \int_t^b g(s)\, ds - g(b) \right\| = \left\| \frac{1}{b - t} \int_t^b (g(s) - g(b))\, ds \right\| \le \frac{1}{b - t} \int_t^b \varepsilon\, ds = \varepsilon.$$

Hence the limit equals $g(b)$. □

Proposition 7.3 (Endpoint verification: ODE holds at $t = \beta$)**.** *Let $D \subset \mathbb{R} \times \mathbb{R}^n$ be open. Suppose $f : D \to \mathbb{R}^n$ is continuous and locally Lipschitz in x. Let $x : (t_0, \beta) \to \mathbb{R}^n$ be a classical solution of*

$$x'(t) = f(t, x(t)) \quad \text{on } (t_0, \beta),$$

assume the (finite) limit $x_\beta := \lim_{t \uparrow \beta} x(t)$ exists, and that $(\beta, x_\beta) \in D$. Let $y : [\beta, \beta + \varepsilon] \to \mathbb{R}^n$ be the (Picard–Lindelöf) solution with $y(\beta) = x_\beta$ and $y'(t) = f(t, y(t))$ for $t \in [\beta, \beta + \varepsilon]$. Define the glued function

$$X(t) := \begin{cases} x(t), & t < \beta, \\ y(t), & t \ge \beta. \end{cases}$$

Then X is C^1 on $[t_0, \beta + \varepsilon]$ and satisfies

$$X'(\beta) = f(\beta, X(\beta)), \qquad X'(t) = f(t, X(t)) \text{ for all } t \in [t_0, \beta + \varepsilon].$$

Proof. (1) Continuity at β. By assumption $x(t) \to x_\beta$ as $t \uparrow \beta$ and $y(\beta) = x_\beta$, so X is continuous at β.

(2) Left derivative at β. For $t < \beta$ the integral form gives

$$X(\beta) - X(t) = x(\beta) - x(t) = \int_t^\beta f(s, x(s))\, ds.$$

Divide by $\beta - t$ and apply Lemma 7.2 to $g(s) = f(s, x(s))$. Since $x(t) \to x_\beta$ and f is continuous at (β, x_β), we have $g(s) \to f(\beta, x_\beta)$ as $s \uparrow \beta$. Hence

$$X'_-(\beta) := \lim_{t \uparrow \beta} \frac{X(\beta) - X(t)}{\beta - t} = \lim_{t \uparrow \beta} \frac{1}{\beta - t} \int_t^\beta f(s, x(s))\, ds = f(\beta, x_\beta).$$

(3) Right derivative at β. Since y is a classical solution,

$$X'_+(\beta) = y'(\beta) = f(\beta, y(\beta)) = f(\beta, x_\beta).$$

(4) Differentiability and the ODE at β. The one-sided derivatives exist and coincide, so X' exists at β and $X'(\beta) = f(\beta, x_\beta) = f(\beta, X(\beta))$. Away from β, X equals either x or y, both classical solutions, hence $X'(t) = f(t, X(t))$ for all $t \ne \beta$. Therefore X is C^1 on $[t_0, \beta + \varepsilon]$ and satisfies the ODE everywhere on that interval, including $t = \beta$. □

Remark 7.4 (What is essential)**.** The argument needs: (i) $x(t) \to x_\beta$ as $t \uparrow \beta$; (ii) $(\beta, x_\beta) \in D$ and f continuous at that point; (iii) a right-hand local solution at (β, x_β) (ensured, e.g., by local Lipschitz in x). Under these, the integral identity plus Lemma 7.2 forces the left derivative to be $f(\beta, x_\beta)$, and the glued right solution forces the right derivative to match.

Theorem 7.5 (Maximal solution and blow-up/escape alternative)**.** *Assume f is continuous and locally Lipschitz in x on D. For any initial data $(t_0, x_0) \in D$ there exists a unique* maximal solution $x : (\alpha, \beta) \to \mathbb{R}^n$ *to* (1) *(i.e. it cannot be extended while remaining in D). If $\beta < \infty$, then either*

(i) $\|x(t)\| \to \infty$ as $t \uparrow \beta$ (blow-up), or
(ii) $\operatorname{dist}((t, x(t)), \partial D) \to 0$ as $t \uparrow \beta$ (escape to the boundary of the domain).

A symmetric statement holds as $t \downarrow \alpha$.

Proof. Existence and uniqueness of a local solution and the possibility of extending as long as the solution remains in a compact subset of D follow from Theorem 4.1 and Lemma 7.1. Define β as the supremum of times to which the solution can be extended; Zorn's lemma is not needed because we can extend step-by-step while the trajectory stays in compact subsets of D.

If $\beta < \infty$ and neither (i) nor (ii) occurs, then there is a compact $K \subset \mathbb{R}^n$ and $\delta > 0$ such that $\{(t, x(t)) : t \in [\beta - \delta, \beta)\} \subset [\beta - \delta, \beta] \times K \subset D$. By Lemma 7.1 we can extend beyond β, contradicting maximality. Hence (i) or (ii) must occur. □

Proposition 7.6 (Global existence under linear growth)**.** *Suppose f is globally Lipschitz in x with constant L and satisfies the linear growth bound*

$$\|f(t, x)\| \le a + b\,\|x\| \qquad \textit{for all } (t, x) \in \mathbb{R} \times \mathbb{R}^n,$$

with $a, b \ge 0$. Then every solution exists for all $t \in \mathbb{R}$ and obeys the a priori bound

$$\|x(t)\| \;\le\; \|x_0\|\, e^{b|t-t_0|} + \frac{a}{b}\Big(e^{b|t-t_0|} - 1\Big) \quad (\textit{interpret } a/b = 0 \textit{ if } b = 0).$$

Proof. From the integral equation,

$$\|x(t)\| \le \|x_0\| + \int_{t_0}^{t} (a + b\,\|x(s)\|\,)\, ds.$$

Let $y(t) := \|x(t)\| + \frac{a}{b}$ (or $y = \|x\|$ if $b = 0$). Then

$$y(t) \le y(t_0) + b \int_{t_0}^{t} y(s)\, ds.$$

Apply Lemma 3.1 with $A = y(t_0)$ and $B \equiv b$ to get $y(t) \le y(t_0) e^{b|t-t_0|}$, which yields the stated bound. The bound precludes finite-time blow-up, so by Theorem 7.5 the solution is global. □

7.1. Osgood blow-up criterion for scalar autonomous ODEs.

Proposition 7.7 (Osgood blow-up (scalar autonomous))**.** *Consider $x' = g(x)$ on $\mathbb{R}$ with $g(x) > 0$ for $x \ge x_0$. Then the (forward) time to reach $+\infty$ is*

$$T_\infty = \int_{x_0}^{\infty} \frac{ds}{g(s)}.$$

If $T_\infty < \infty$, the solution blows up at finite time $t_0 + T_\infty$. If $T_\infty = \infty$, no forward blow-up occurs.

Proof. By separation of variables,

$$\int_{x_0}^{x(t)} \frac{ds}{g(s)} = t - t_0.$$

As $x(t) \uparrow \infty$, the left-hand side tends to T_∞. Thus x reaches $+\infty$ at time $t_0 + T_\infty$ if and only if $T_\infty < \infty$. □

8. Linear systems as a corollary

Proposition 8.1 (Linear systems). *Let $A : [a, b] \to \mathbb{R}^{n\times n}$ and $b : [a, b] \to \mathbb{R}^n$ be continuous. The IVP*

$$x'(t) = A(t)x(t) + b(t), \qquad x(t_0) = x_0, \quad t_0 \in [a, b],$$

has a unique solution on $[a, b]$. Moreover, if $L := \sup_{t\in[a,b]} \|A(t)\|$, then

$$\|x(t) - y(t)\| \le e^{L|t-t_0|} \|x_0 - y_0\|$$

for any two solutions x, y with possibly different initial data.

Proof. Set $f(t, x) = A(t)x + b(t)$. Continuity on $[a, b] \times \mathbb{R}^n$ is clear, and $\|f(t, x) - f(t, y)\| = \|A(t)(x - y)\| \le L \|x - y\|$ gives Lipschitz in x on $[a, b]$. Apply Theorem 4.1 and Proposition 4.3. □

9. Worked examples (uniqueness / blow-up behavior)

Example 9.1 (Logistic). $x' = rx(1 - x/K)$, $x(0) = x_0$, $r, K > 0$. Here $f(x)$ is locally Lipschitz and satisfies linear growth, so existence is global and uniqueness holds. The explicit solution is

$$x(t) = \frac{Kx_0e^{rt}}{K + x_0(e^{rt} - 1)}.$$

Example 9.2 (Finite-time blow-up but unique until blow-up). $x' = x^2$, $x(0) = x_0 > 0$. Solution $x(t) = \dfrac{x_0}{1 - tx_0}$ blows up at $t^* = 1/x_0$. Local Lipschitz gives uniqueness on $(-\infty, t^*)$.

Example 9.3 (Non-Lipschitz with uniqueness (Osgood)). $x' = x \log(\frac{1}{|x|})$ near $x = 0$, $x(0) = 0$. Differences satisfy $\|f(x) - f(y)\| \lesssim \|x - y\| \log(\frac{1}{\|x-y\|})$, and $\int_0 \frac{dr}{r \log(1/r)} = \infty$; Osgood yields uniqueness.

Example 9.4 (Non-uniqueness (Peano)). $x' = \sqrt{|x|}$, $x(0) = 0$. Osgood fails ($\int_0 r^{-1/2}dr < \infty$), and there are infinitely many solutions: $x \equiv 0$ and "delay-start" parabolas $x(t) = \frac{1}{4}(t - \tau)^2$ for $t \ge \tau \ge 0$.

Summary of hypotheses vs. conclusions

Hypothesis on f	Existence	Uniqueness	Maximal interval characterization
Continuous	✓ (Peano)	×	Maximal solution; ends by blow-up/escap
Lipschitz in x (local)	✓	✓	Maximal solution; blow-up/escape alterna
One-sided Lipschitz	✓	✓	As above
Osgood modulus $\int_0 \frac{dr}{\omega(r)} = \infty$	✓(via local existence)	✓	As above
Global Lipschitz + linear growth	✓(global)	✓	No finite-time blow-up